MILITARY SPECIFICATION

DETECTOR, PIN, FIBER OPTIC,

820-910 NANOMETERS WAVELENGTH RANGE,

GLASS-PIGTAILED TYPE (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for positive-intrinsic-negative (PIN) fiber optic, glass-pigtailed detectors.

1.2 Classification. Detector classification shall be in accordance with DOD-D-24620.

TABLE I. Detector characteristics and part numbers.

Military part number D 24620/1-	Sealing level	Wavelength of operation	Pigtail fiber	
			Core/cladding diameter (µm)	Core index profile
A 08 A B 08 A	hermetic nonhermetic	820-910 nanometers	50/125	step
A 08 B B 08 B	"	"	"	graded
A 08 C B 08 C	"	"	100/140	step
A 08 D B 08 D	"	"	"	graded

NOTE The detector covers the wavelength band from 820 and 910 nanometers, but its performance criteria shall be met at the specified center wavelength. The center wavelength specified is the center operating wavelength of the LED source as specified in DOD-S-24622.

1.2.1 Part number. The part number shall be in accordance with DOD-D-24620 (see table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command (SEA 55Z3), DOD Standardization Program and Documents Division, Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Design maximum and minimum ratings.

Minimum continuous forward current (I_{FM}) at T_A = 25°C - - - - - - - - 10 mA

(Derate I_{FM} above T_A = 50°C at 0.2 mA/°C)

Minimum peak forward current (I_{FP}) at T_A = 25°C, t_p = 1 µs, DF = 0.1 - - 100 mA

Minimum power dissipation (P_{DM}) at T_A = 25°C- - - - - - - - - - - - - - 15 mW

Minimum dc reverse voltage (V_{RM}) at T_A = 25°C - - - - - - - - - - - - - 100 V

Storage and operating temperature range:

Hermetic pigtailed PINS (Quality level A):

(operating): T_C = -55°C to +105°C.

(storage): T_C = -55°C to +105°C.

Nonhermetic pigtailed PINS (Quality level B):

(operating): T_C = -55°C to +105°C.

(storage): T_C = -55°C to +105°C.

External lead maximum solder temperature (5 seconds)- - - - - - - - - - 250°C

Pigtail assembly, minimum bend radius - - - - - - - - - - - - - - - - - - - 10 mm

Pigtail assembly, pull strength (minimum) - - - - - - - - - - - - - - - - - 20 N

Pigtail fiber, pull strength (minimum)- - - - - - - - - - - - - - - - - - - 5 N

2. APPLICABLE DOCUMENTS

2.1. Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-19491 - Semiconductor Devices, Packaging of
MIL-S-19500 - Semiconductor Devices, General Specification for
DOD-D-24620 - Detectors, Pin, Fiber Optic, General Specification for
DOD-S-24622 - Sources, LED, Fiber Optic, General Specification for

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-750 - Test Methods for Semiconductor Devices
MIL-STD-889 - Dissimilar Metals

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of conflict between the text of the specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with DOD-D-24620, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations, symbols, and definitions used herein shall be as specified in DOD-D-24620 and its appendix.

3.3 Design, construction and physical dimensions. Pins shall be of the design, construction and physical dimensions as specified on figure 1.

3.3.1 Lead conductors.

3.3.1.1 Material and finish. Anode and cathode lead conductors shall be Alloy-52 or Kovar of diameter 0.51 millimeter (mm) (American Wire Gauge 24) with a gold plated, hot solder dip finish. Conductors shall be diametrically opposed and displaced an equal distance from the pin assembly center-line axis (see symbol M, figure 1, for the conductor spacing).

3.3.1.2 Polarity identification. A black stripe, 1 mm wide and 13.2 mm long (see dimension symbol C, figure 1) shall be affixed to the assembly case, collateral and parallel with the cathode conductor.

3.3.2 LED case assembly. The pin case and stud assembly shall be metallic and gold plated with a visually smooth surface finish. The assembly shall be noncorrosive and meet the requirements of MIL-STD-889. Anode and cathode leads shall be electrically isolated from the case.

3.3.3 Pigtail assembly.

3.3.3.1 Sheath. The fiber pigtail sheath material and construction shall provide the bend-restrictive property which prevents the pigtail assembly from exceeding the minimum bend radius specified in 1.3. Sheath materials shall be nonmetallic.

3.3.3.2 Strain relief. Strain relief shall be provided between the case body and the pigtail sheath. Metallic or nonmetallic materials may be used for strain relief construction. The overall length of the strain relief member shall not exceed 70 mm and shall provide a minimum bend radius of 20 mm, under normal usage.

3.3.3.3 Fiber. The pigtail fiber shall contain a thin protective coating easily removed without damage to the fiber by mechanical or chemical means. The fiber termination shall be unconnectorized.

3.4 Performance characteristics. Specific performance levels required of each detector type to satisfy the burn-in and qualification inspections' requirements are as noted in tables I, II, III, and IV.

3.5 Color. Pin diode package color shall be gold. Pigtail exterior jacketing shall be light blue for 50/125 micrometer (core/cladding) fibers, and light red for 100/140 micrometer (core/cladding) fiber sizes.

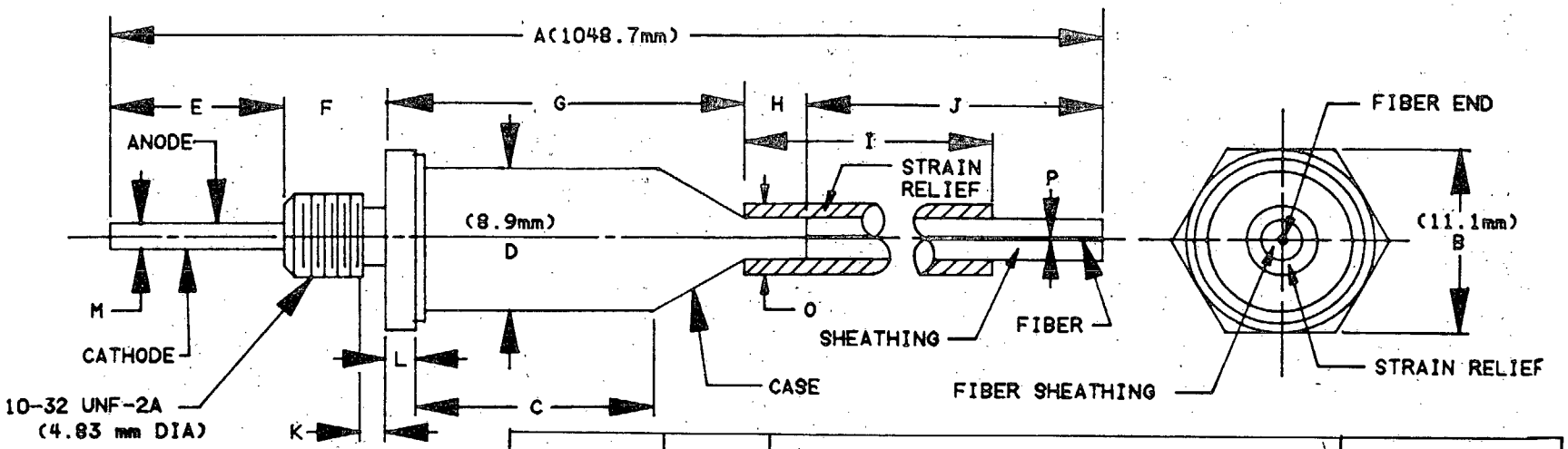
3.6 Mass. The detector mass (pigtail included) shall be a minimum consistent with the packaging design employed, and shall not exceed 10 grams.

3.7 Marking.

3.7.1 Identification. Marking size for detector type-designation shall be 3 mm in height, and the manufacturer's designating symbol shall be identifiable without need for visual magnification.

3.8 Workmanship. Detectors shall show no evidence of excessive stress in the pigtail/strain relief region such as poor attachment, crimp or pinch conditions.

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Detector type	Letter symbol	Dimensions				Notes 1 and 5 applies to all
		Millimeters		Inches		
		Min	Max	Min	Max	
All types (see 1.2)	A	1035.1	1060.5	40.75	41.75	
	B	10.9	12.2	.43	.48	
	C	13.0	13.5	.51	.53	
	D	8.4	9.0	.33	.35	
	E	13.0	13.5	.51	.53	
	F	7.4	7.9	.29	.31	
	G	19.8	20.3	.78	.80	
	H	7.6	7.9	.30	.31	
	I	69.4	70.6	2.73	2.78	2
	J	990.6	1009.7	39.00	39.75	3
Sealing levels A and B	K	1.17	1.19	.046	.047	
	L	1.96	2.01	.077	.079	
	M	1.96	2.01	.077	.079	4
	O	4.8	5.1	.19	.20	
	P	0.13/0.15		.005/.006		6

NOTES:
1. Inch equivalents are given for general information only.
2. Dimension I is a maximum value (see 3.3.3.2).
3. Dimension J is the overall pigtail fiber length from the end of the detector case to the pigtail end.
4. Dimension M is the center-to-center spacing between anode and cathode conductors.
5. Tolerance does not apply to dimension P.
6. Dimension P shall be in accordance with the type classification specified in 1.2.
7. Dimensions are in millimeters.

FIGURE 1. Pin detector package assembly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. Quality assurance provisions shall be in accordance with DOD-D-24620, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the inspections specified in table II (screening) and tables III, IV and V.

4.2.1 Screening inspection. Process conditioning and screening inspection shall be conducted on 100 percent of the lot prior to submission of the lot to the tests specified in tables III, IV, and V herein. Table II lists the parameters with their interim and final end-point limits (for steady-state reverse voltage burn-in) which must be met for detector acceptance. Tests shall be accomplished in the sequence shown in table II. End-point parameters shown in table II are either of the maximum limit type or of the variational (delta, Δ) type. In particular, the pretest burn-in parameters shown in table II represent their range of values as specified for pretest detector acceptance. The Δ parameter values shown for post-test burn-in screening represent the variation in any particular parameter for post-test part acceptance. Parameter values falling outside these absolute and delta limits shall be cause for detector rejection.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the inspection tests specified in tables III, IV, and V.

4.3.1 Group A inspection. Group A inspection tests, LTPD's and performance limits are specified in table III.

4.3.2 Group B inspection. Group B inspection tests, LTPD's and performance limits are specified in table IV. Required end-point measurements are as specified following subgroups 2 and 3.

4.3.3 Group C inspection. Group C inspection tests, LTPD's and performance limits are specified in table V. Required end-point measurements are as specified following subgroups 2, 3 and 5.

TABLE II. Process conditioning and screening inspection.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
High temperature life (nonoperating)	1031	Time = 24 hrs. T_A = 105°C (QLA) T_A = 105°C (QLB)	---	---	---	---
Thermal shock (temperature cycling)	1051 (107 of MIL-STD-202) Test condition A-1	$T_{A(min)}$ = -55°C (QLA/QLB) $T_{A(max)}$ = 105°C (QLA) = 105°C (QLB) No special mounting conditions.	---	---	---	---
Constant acceleration	2006	20,000 G, 6 directions	---	---	---	---
Steady-state reverse voltage (burn-in) Pre burn-in (interim) end points:		V_R = -45 V. T_A = 25°C. λ_p = 820, 850, 880 or 910 nm, as specified $\Delta\lambda$ = 40 ±5 nm				
Capacitance	4001	ø = 0 W F = 100 kHz	C	1	3	pF
- - -						

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TABLE II. Process conditioning and screening inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Details	Symbol	Limits Min	Limits Max	Unit
Reverse current at reverse bias voltage (dark current)	3061	ϕ = 0 W	I_d	3	7	nA
Responsivity	---	f_m = 10 kHz λ_1 = 820 nm, λ_2 = 850 nm, λ_3 = 910 nm, ϕ_1 = 30 nW ϕ_2 = 10 µW, ϕ_3 = 300 µW	R	0.45 for each specified λ_p	0.55	A/W
Responsivity linearity	---	Calculated from R data taken above.	R_L	-10	+10	%
Spectral responsivity and quantum efficiency	---	Calculated from R data taken above.	n	65	75	%
Noise equivalent power	---	f_m = 10 kHz calculated from the I_d and R data taken above.	NEP	$5x10^{-13}$	$5x10^{-14}$	$W/\sqrt{Hz}$
Pulse rise time	---	PRF = 10 MBPS D.F. = 50%	t_r	2	4	ns
Pulse fall time	---	PRF = 10 MBPS D.F. = 50%	t_f	2	4	ns
Analog bandwidth	---	3dB BW = 10 MHz	BW	10	-	mHz
Burn-in	1038 Test condition A	Time = 48 hrs. V_R = -80 V ϕ = 500 µW T_A = +105°C (QLA) T_A = +105°C (QLB)				
Post burn-in (final) end points:		Same as pre burn-in end points				
Capacitance	4001	Same detail conditions	ΔC	-	+10	%
Dark current	3061	as noted under pre burn-in	ΔI_d	-	+10	%
Responsivity	---	"	ΔR	-5	+5	%
Responsivity linearity	---	"	ΔR_L	-10	+10	%
Spectral responsivity and quantum efficiency	---	"	Δn	-5	-	%
Noise equivalent power	---	"	ΔNEP	-	5	%
Pulse rise time	---	"	Δt_r	-	10	%
Pulse fall time	---	"	Δt_f	-	10	%
Analog bandwidth	---	"	ΔBW	10	-	MHz
Hermetic seal	1071	Leak > 10^{-8} atm cc/s (QLA only)	---	---	---	---
Fine leakage rate	Test condition H					

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TABLE II. Process conditioning and screening inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Details	Symbol	Limits Min	Limits Max	Unit
Gross leakage rate	Test condition C	Leak > 10^{-3} atm. cc/s (QLB only) T_{FLUID} = +105°C	---	---	---	---
Visual and mechanical inspection	2071	Pigtail attachment	---	---	---	---

TABLE III. Group A inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Details	LTPD	Symbol	Limits Min	Limits Max	Unit
Subgroup 1							
Visual and mechanical inspection	2071		5	---	---	---	---
Subgroup 2		T_A = +25°C, ø = OW					
Capacitance	4001	f = 100 kHz		C	1	3	pF
Dark current	3061	V_R = -45 V	5	I_d	3	7	nA
Reverse current leakage and breakdown voltage	4021	V_R = -45 V		I_R	-	10	nA
Subgroup 3		T_{min} = -55°C					
Breakdown voltage	4021	T_{max} = +105°C (QLA) = +105°C (QLB)	5	I_R	-	10	µA
Dark current	3061	V_R = -100 V V_R = -45 V ø = OW		I_d	-	10	nA
Subgroup 4		T_A = +25°C					
Responsivity	---	f_m = 10 kHz		R	0.45	0.55	A/W
Noise equivalent power	---	f_m = 10 kHz	5	NEP	$5x10^{-13}$	$5x10^{-14}$	$W/\sqrt{Hz}$
Pulse rise time	---	PRF = 10MBPS, D.F. = .5		t_r	2	4	ns
Pulse fall time	---	PRF = 10MBPS, D.F. = .5		t_f	2	4	ns
Analog bandwidth	---	3 dB BW = 10 MHz		BW	10	-	MHz

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TABLE IV. Group B inspection.

Inspection	MIL-STD-750		LTPD
	Method	Details	
Subgroup 1			
Solderability Resistance to solvents	2026 1022		15
Subgroup 2			
Thermal shock (temperature cycling)	1051	$T_{A(min)}$ = -55°C $T_{A(max)}$ = +105°C (QLA) = +105°C (QLB) No special mounting	
Hermetic seal Fine leakage rate	1071 Test condition H	QLA detectors only	10
Gross leakage rate	Test condition C	QLB detectors only	
End points: Same as subgroup 2 of group A inspection.		Detectors tested only to gross leak step 1 procedure with fluid temp. = +105°C	
Subgroup 3			
High temperature (nonoperating) Life (LTPD)	1032	Time = 24 hrs. T_A = +105°C (QLA) = +105°C (QLB)	10
Subgroup 4			
Burn-in	1038 Test condition A	V_R = -80 V Time = 48 hrs. ø = 500 µW T_A = +105°C (QLA) T_A = +105°C (QLB)	5
End points: Same as subgroup 2 of group B inspection plus subgroup 4 of group A inspection.			
Subgroup 5			
Decap internal visual design verification Bond strength	2075 2037	Test condition A	15

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TABLE V. Group C inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Details	LTPD
Subgroup 1			
Physical dimensions	2066		15
Subgroup 2			
Thermal shock (glass strain)	1056	T_{FLUID} = +105°C (QLA) = +105°C (QLB)	
Terminal strength Tension	2036 Test condition A	Electrical leads Tensile load = 5 Newtons for 5 minutes	10
		Fiber pigtail assembly Tensile load = 20 Newtons for 5 minutes	
Torque	Test condition D1	Electrical leads Torque load = 0.1 Newton-meter (1 lb-inch) for 10 seconds	10
		Fiber pigtail assembly Torque load = 0.2 Newton-meter (2 lb-inch) for 10 seconds	
Hermetic seal	1071		
Fine leakage rate	Test condition H	QLA detectors only	
Gross leakage rate	Test condition C	QLB detectors only Detectors tested only to gross leak step 1 procedure with fluid temp. = +105°C	
Moisture resistance	1021		
Visual and mechanical inspection	2071		
End points: Same as subgroup 2 of group A inspection	---		---
Subgroup 3			
Shock Vibration (variable freq.) Constant acceleration	2016 2056 2006		10
End points: Same as subgroup 2			

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TABLE V. Group C inspection - Continued.

Inspection	MIL-STD-750		LTPD
	Method	Details	
Subgroup 4			
Salt atmosphere (corrosion)	1041		15
Subgroup 5			
Burn-in	1038 Test condition A	T_A = +105°C (QLA) = +105°C (QLB) V_R = -80 V O = 500 µW Time = 1000 hrs. normal stud mounting.	5
End points: Same as subgroup 2 plus subgroup 4 of group A inspection.			

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19491.

6. NOTES

6.1 Notes. In addition to the notes specified in DOD-D-24620, the following are applicable to this specification.

6.2 Ordering data. Acquisition documents should specify the following:

a. Lead length, material, finish if other than that specified, or when a choice is required by the detector application.
b. Pigtail fiber assembly length if other than that specified, or when a choice is required by the detector application.
c. Mounting arrangement to be used when unique to the intended detector application.

6.3 Engineering data. Electrical and optical operating characteristics contained in table VI are provided as engineering data.

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TABLE VI. Primary pin detector performance characteristics.

Pin Designation	CHARACTERISTICS 14/												
	Pigtail			Electrical			Optical				E/O System		
	Core Index 1/	N/A 2/	Fiber Size 3/	V_R (V) 4/	C (pF) 5/	I_d (NA) 6/	ϕ_R (dB_m) 7/	R (A/W) 8/	η (%) 9/	NEP (W/ $\sqrt{HZ}$) 10/	T_r (NSe) 11/	BW (MHz) 12/	L_R (%) 13/
A08A, B08A	S	0.30	50/ 125	-45	2	5	-5 To -45	0.5	70	10^{-13}	3	10	±10
A08B, B08B	G	0.15											
A08C, B08C	S	0.30	100/ 140										
A08D, B08D	G	0.15											

NOTES:

1. Core index types S and G identify step and graded core index profiles, respectively.
2. NA specifies the nominal numerical aperture of the pigtail fiber.
3. Fiber size is given as the nominal core diameter over the nominal overall fiber diameter for the pigtail fiber.
4. V_R is the specified reverse bias voltage for the PIN in volts.
5. C is the PIN junction capacitance in picofarads.
6. I_d is the PIN dark current in nanoamperes.
7. ϕ_R is the minimum specified input power rage of the PIN.
8. R is the PIN responsivity in amperes per watt.
9. η is the PIN quantum efficiency in percent.
10. NEP is the PIN noise-equivalent-power in watts per root hertz.
11. t_r is the PIN rise time in nanoseconds.
12. BW is the minimum acceptable bandwidth in megahertz.
13. L_R is the minimum acceptable responsivity linearity over the specified power range (ϕ_R) in percent.
14. The above performance parameters are specified at an ambient temperature, T_A, of 25°C.

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Army - CR
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-D-24620/1	Detector, Pin, Fiber Optic, 820-910 Nanometers Wavelength Range, Glass-Pigtailed Type (Metric)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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